Dr James Hardwick

EMPLOYMENT HISTORY

August 2025 – August 2026: **University of Stirling – Lecturer in Earth Science.** *Key responsibilities:* Development and provision of teaching across all stages of undergraduate degree programs in environmental and ecological sciences. Alongside teaching, co-investigator on the Par Forestry funded project optimising deployment of river restoration actions to maximize environmental benefits in northeast Scotland.

May 2025 – August 2025: **Newcastle University – Research Assistant on the internally funded project 'Impact and engagement of climate adaption and resilience in the Kilombero Landscape, Tanzania.'** *Key responsibilities:* Development of coupled hydrological-biophysical model to track water flow, soil nutrients distribution and sediment loss; develop and deliver a short teaching course for in-country stakeholders including university academics and students for capacity building.

October 2024- March 2025: **University of Stirling – Research Assistant on the GERF-ISPF funded project 'Climate Adaptation and Resilience: Rural agricultural communities and industries along the Kilombero River floodplain landscape, Tanzania.'** *Key responsibilities:* Co-project managing tropical catchment research; coordinating & carrying out international fieldwork; in-country capacity building; data management, modelling and analysis; first authored publication currently in prep.

December 2023 – March 2024: **Newcastle University – Research Assistant on the internally funded project 'Exploring understanding and experience of responsibilities for marginalised fieldworkers' safety and wellbeing.'** *Key responsibilities:* Systematic collation of fieldwork policies, risk policies and risk assessments across Newcastle University; document coding and data extraction; led groups in participatory workshops; Co-authored publication currently in prep.

April 2023-December 2023: **Newcastle University – Research Assistant on the internally funded 'Assessing UK higher education institutions policy for fieldwork safety and inclusion.'** *Key responsibilities:* Systematic collation of UK higher education institutional fieldwork policies and risk assessments; document coding and data extraction. Co-authored publication currently under review at *Ecological Solutions and Evidence*.

EDUCATION

2022-2025: **PhD in Biology, School of Natural and Environmental Sciences, Newcastle University**, internally funded by an internal Newcastle University studentship.

Title: The fluvial biogeomorphic impacts and feedbacks of riparian invasive non-native plant species.
Supervisors: Dr Zarah Pattison, Dr Christopher Hackney, Professor Clare Fitzsimmons

2017-2021: **MGeol (Hons.) Geology (First Class), School of Geography, Geology and the Environment, University of Leicester**

- Master's thesis: Disentangling the Influence of Invasive Plant Species on River Morphology.

TEACHING

I am currently working towards Associate Fellow of the Advance Higher Education. I have a range of teaching experience including module coordination, residential field course group teaching, workshop demonstrating, supervising undergraduate dissertation students, the training of research field assistants and assessments – including formative and summative assessment.

Stirling University (Lecturer)

2026 ENVU6EA (Stage 3, Methods and Applications in Environmental Sciences) Work based learning with active client, Water of Leith Conservation Trust, to conduct environmental monitoring of water quality.

2026 GEOU4ER (Stage 2, Environmental Resilience) Introduced concepts around fluvial resilience and how fluvial systems can be modelled to achieve targeted outcomes.

2026 GEOU2GE (Stage 1, Global Environmental Issues) Introduce sand mining and sedimentological degradation of riparian habitats.

2025 ENVU1BP (Stage 1, Building Planet Earth) Introduced a range of core geological concepts, developed field-based skills including field sketching and rock identification

2025 GEOU1PP (Stage 1, People and the Environment) Introduced the anthropogenic and biodiversity concepts.

2025 SCIU3LS (Stage 3, laboratory skills for environmental science) Taught laboratory based practical classes using a range of analytical techniques and seminars, covering a wide range of applied environmental chemistry.

2025 SCIU1FI (Stage 1, Fundamental of Science Inquiry) Supervised group-based field work centered around developing research strategy and hypothesis testing to understand how decomposition rates varied across different environmental settings. Provided both formative and summative feedback for the students assessed work.

Newcastle University (PhD demonstrator)
Residential field course teaching:
2022 BIO2032 (Stage 2 Residential field course, *Northumberland*) Taught freshwater macro-invertebrate sampling techniques and identification. Supervised group projects using field collected data to assess the role of different land cover, water chemistry and mining activity on invertebrate community diversity. I supervised students in data analysis and report writing and undertook the marking and provision of formative and summative feedback on their assessed work.

2023 BIO2040 (Stage 2 Residential field course, *Scotland*) Taught vegetation identification and surveying techniques. Supervised group projects using field data to assess the environmental drivers of plant community composition. Provided both formative and summative feedback for students' assessed work.

Workshop teaching:
2022 - NES8313 (MSc, Coupled Human and Natural Systems) Supervised group-based discussions with input from my personal expertise to aid student understanding of the links between human and natural systems. Provided summative feedback on exams.

2022 BIO3039 (Stage 3 Biodiversity Science and Management) Supervised group-based workshop and presentations on a range of topics from land conservation to managing habitats to alleviate human impacts. I also provided summative feedback for group presentations.

Practical sessions:
2022 BIO1021 (Stage 1 Diversity of Life: Form and Function) Led lab-based practical sessions on anatomical descriptions of invertebrate specimens and phylogenies. I also marked and provided summative feedback on students' practical exams.

Undergraduate dissertation co-supervision:
2024 – 2025 Abigail Cato (University of Stirling) *Using minirhizotrons to explore the impact of Impatiens glandulifera invasion on riparian root characteristics*

RESEARCH SKILLS AND EXPERIENCE

Fieldwork: Extensive experience in diverse range of landscapes and ecosystems, including temperate woodlands, tropical rainforests and freshwater habitats. Experience in plant community surveying, water quality testing, sediment sampling, geotechnical measurements, geological lithological, structural mapping and sensor (e.g. water level and turbidity) deployment.

2025 – present	Water of Dye, Northeast Scotland
2022-25	Fieldwork to Kilombero Valley, Tanzania
2022-25	Repeated fieldwork across central Scotland and Northen England (PhD data collection)
2019	Independent mapping project (1 month) on Isle of Raasay
2017-2021	Structural and lithological geological mapping training (North Wales, Northwest Scotland, Peak District, Southern Spain)

Data processing and analysis: Advanced data manipulation and visualisation of large datasets in R. Proficiency in advanced statistical modelling techniques in R, including LMM, NMDS and Structural Equation Modelling, at a range of spatial and temporal scales and on a range of data **(**multivariate, species diversity, environmental). Experience with remotely sensed data in QGIS, and ArcGIS Pro and process-based earth systems models in C# and Python such as CAESAR-Lisflood, SWAT+ and Landlab.

Laboratory: Experience with setting up and running hydrodynamic experiments to assess grains fluvial behavior, isotopic analysis using mass spectrometry to track environmental change, environmental chemical analysis, water chemistry testing, grain size analysis, anatomical dissection, rock thin section preparation and field-based sensor development.

Project management: Skilled in managing and coordinating research projects, including planning and executing domestic and international fieldwork. Excellent time management and outstanding organisational skills, with adaptable communication to a range of technical and non-technical audiences. Experience with managing budgets for research projects. Regularly use version control systems, GIT, for collaborative research and code management.

People management: Experience with stakeholder collaboration and dissemination, including landowners, businesses, practitioners, policy groups, citizen scientists and volunteers.

FUNDING APPLICATIONS

2026	**Co-I,** Par Forestry IV holdings limited funded project (**£44,000**) **'***Optimising Deployment of River Restoration Actions to Maximise Environmental Benefits*' in northeast Scotland.
2024	**PI**, British Society for Geomorphology Postgraduate Research Grant (**£1020**) to fund a pilot study '*Roots and riverbank stability: effects of riparian non-native plant invasions on bank stability'*.
2024	**Co-I**, Scottish Funding Council through the UK Department for Science, Innovation and Technology international sciences partnerships fund (GERF-ISPF) (**£10,000**) grant for the project 'Climate Adaptation and Resilience: Rural agricultural communities and industries along the Kilombero River floodplains landscape'.
2023	**Co-I,** International Union for Conservation of Nature contract (**£3000**) to produce a Management, Measures and Cost Note for the invasive macrophyte *Crassula helmsii* (New Zealand Pigmyweed), on behalf of the European Union Commission.
2022	**PI**, Newcastle University Doctoral College Fund (**£1900**) to fund fieldwork in Tanzania, investigating the role sugar cane farming plays on catchment erosional processes.

PUBLICATIONS

ORCID ID: 0000-0002-4431-009X

Accepted:

1. Keen, E., **Hardwick, J.,** Novoa, A., Gómez, R., Willby, N., Mill, A., Pattison, Z. (*2026*). Insights into the management of invasive alien plants in forests: a global perspective. *Journal of Applied Ecology.*
2. **Hardwick, J.,** Hackney, C., Law, A. Pattison, Z. (2026). Invasive non-native plants indirectly destabilise riverbanks. *Biological Invasions*. https://doi.org/10.1007/s10530-025-03721-2
3. Mair, L., **Hardwick, J.,** Mannion, N., Braunholtz, L., Carlin, B., Hopkins, P., Sikka, T., Pattison, Z. (*2025*). Improving university policies and risk assessment to support inclusive fieldwork. *Ecological Solutions and*

Evidence. https://doi.org/10.1002/2688-8319.70109

4. **Hardwick, J.**, Hackney, C., Keen, L., Fitzsimmons, C., Willby, N., Pattison, Z. (2025). The role of non-native plant species in modulating riverbank erosion: a systematic review. *River Research and Applications*. https://doi.org/10.1002/rra.4420

Submitted:

1. Mair, L., Jackson, K., **Hardwick, J.,** Mannion, N., Hopkins, P., Howes, K., Maddison, J., O'Donnel, A., Scott, S., Sikka, T., Pattison, Z. (*revision submitted*). Exploring understanding and experience of responsibilities for marginalised fieldworkers' safety and wellbeing with recommendations for more inclusive approaches. *Journal of academic ethics*.

In preparation (<3 months until submission):

1. **Hardwick, J.,** Hackney, C., Pattison, Z. Evaluating the direct and indirect effects of invasive non-native riparian plant invasions on riverbank erosion. To be submitted to *River Research and Applications.*
2. **Hardwick, J.,** Hackney, C., Pattison, Z. Plant invasions alter river channel sediment dynamics along an invasion gradient. To be submitted to *Earth Systems Dynamics*.
3. **Hardwick, J.,** Shirima, D., Pfeifer, M., Seki, H., Davies, S., Pattison, Z. (*in prep*). Assessing the impact of climate change on key agroforestry species distribution across Tanzania, East Africa. To be submitted to *Biotropica*.
4. **Hardwick, J.,** Hackney, C., Shirima, D., Pfifer, M., Pattison, Z. Evaluating the combined impacts of agricultural land use policy and near-future climate change on soil loss across a tropical catchment. To be submitted to *CATENA.*

OTHER WRITING

1. **Hardwick, J.**, Pattison, Z., Galasso G., Brundu, G., Venter, T., Postigo J.L., Nunes, A. (2023) Management Measures and Cost Note: The Management of Cockayne (*Crassula helmsii*). International Union for Conservation of Nature (IUCN).

PRESENTATIONS

Sep 2024	Presentation at Neobiota Conference (Lisbon, Portugal): '*Seed transport and landscape modelling to predict riparian plant invasion hotspots'*
April 2023	Presentation at the Scottish Freshwater Group (Stirling, UK): '*Unravelling the role of a non-native plant species, Impatiens glandulifera (Himalayan balsam), on modulating riverbank erosion'*

Poster presentations: American Geophysical Union annual meeting, San Francisco, USA (Dec, 2023); Symposium for European Freshwater Sciences conference, Newcastle, UK (June, 2023); British Ecological Society conference, Edinburgh, UK (Dec, 2022); British Society for Geomorphology, Newcastle, UK (Sep, 2022); British Sedimentary Research Group, online, (Dec, 2020)

QUALIFICATIONS, AWARDS, MEMBERSHIPS AND OTHER ROLES

Qualifications & Awards

2026 – 2029	REC level 3 Outdoor First Aid Course
2021	East Midland Geological Society Beryl Whittaker Prize award for the highest mark in the MGeol research project (University of Leicester)
2016	Full clean UK driver's license

Professional scientific society memberships

British Ecological Society
American Geophysical Union
British Society of Geomorphology

Committee/volunteer roles

2022-2025	Postgraduate Research Representative at the School of Natural and Environmental Sciences field trip and off campus health and safety committee
2022 -24	Upper Calder Catchment Volunteer Coordinator for the Calder Rivers Trust, macro-invertebrate monitoring program
2017 – 2025	Duke of Edinburgh Gold award group residential field instructor for Hills Road Sixth Form College (UK KS5), Cambridge

COURSES AND TRAINING

Technical: Structural Equation Modelling for Ecologists (PR Statistics, March 2023); Programming with Python (Newcastle University, 2023); Species identification and field mapping (FSC, May 2022)

Teaching: Higher Education teaching practices (Newcastle University, January 2023); Introduction to Teaching and Learning in Higher Education (Newcastle University, April 2022)

REFEREES

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